

A Hybrid Framework for Real-Time Human Coordination and Predictive Maintenance in Multi-Agent Systems

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I. INTRODUCTION

The manufacturing sector underwent a complete transformation because Industry 4.0 merged cyber-physical systems with Internet of Things (IoT) and big data analytics and artificial intelligence (AI) to create smart factories which function as connected networks [1], [2]. The new system design will achieve improved operational performance through its ability to share data instantly and its automatic system selection capability. The core element of this transformation relies on Predictive Maintenance (PdM) which uses sophisticated analytical methods to analyze sensor information for equipment failure prediction to prevent unexpected system failures and lengthen asset operational periods [5]. The high-risk sectors of automotive and aerospace and petrochemicals use PdM to achieve cost reductions of 25–30% while enhancing their equipment performance (OEE) [6], [12]. PdM technology enables the detection of electrode deterioration which occurs because of vibration and temperature changes in resistance spot welding operations used for automotive manufacturing thus avoiding production halts [11], [28]. The challenges of Industry 4.0 require PdM systems to become more resilient because they face three main issues which include data heterogeneity and cyber-physical system vulnerabilities and production demand changes [13], [15]. The traditional PdM methods provide these advantages yet they face challenges when implementing them across multiple machines and they lack human operator knowledge which results in poor decision making during unpredictable situations [3], [7].

II. RELATED WORK

Industry 4.0 manufacturing systems require independent operation through data intelligence and human involvement to achieve success in their complex and dynamic operational environments [1]. The system transformation needs three fundamental elements which consist of Multi-Agent Systems (MAS) for decentralized coordination and Predictive Maintenance (PdM) for reliability and Human-in-the-Loop (HITL) frameworks for contextual judgment. The development of these sys-

tems took place independently between different fields which resulted in fundamental differences between their operational features and their interface accessibility [2, 3]. MAS optimize industrial complexity through distributed, collaborative agents which enable better scalability and adaptability [1, 2] but their autonomous nature tends to ignore essential human operational knowledge [3]. The system uses IoT technology and machine learning to predict equipment breakdowns which decreases operational interruptions [5] but it operates through reactive data-based systems which lack human judgment to handle immediate system changes [6]. HITL systems work to connect these two groups through human feedback which helps AI systems make better decisions while reducing incorrect alerts [7] but they do not have established methods for humans to interact with agents effectively [3, 4]. The review examines the core principle which shows Industry 4.0 adaptive manufacturing faces its primary challenge because its three vital elements function independently from one another. The three pillars need to work together through shared communication systems and adaptive teamwork methods and understandable interface systems to create a manufacturing system which unites artificial intelligence operational excellence with human decision-making abilities and system stability.

A. Multi-Agent Systems in Manufacturing and Industrial Automation

1) *Definitions and Key Characteristics:* MAS represent computational systems which use multiple self-governing agents that possess separate abilities and functions to solve problems which require teamwork because single agents cannot handle them [1]. The agents function independently because they detect their surroundings and use their internal data and outside information to make decisions which lead to action execution for achieving common goals [2]. MAS operate differently from single-agent systems because they use distributed intelligence which enables agents to work together through interaction and coordination instead of depending on central control for system operation [1].

2) *Key characteristics of MAS*: The core principle of decentralized manufacturing operates through independent agents which maintain their internal states while executing actions without any external direction [3]. The inspection cells of UR3 robotic systems in automotive component assembly plants use agents to modify their quality check schedules through sensor data feedback which enables 25% better workflow flexibility than traditional centralized systems [4]. The system faces delays in collaboration when agents receive too much freedom because production agents will keep operating at their default speed when inspection agents cannot share adjustment information [4]. The limitation creates a direct connection to the coordination overhead problem which Section 3.3 discusses. *Social Ability*: Agents use FIPA-ACL standardized protocols to exchange information while they distribute resources and resolve conflicts [1, 6]. The system requires this functionality to operate correctly during changes in operational conditions. The Modbus protocol enables material handling agents (AGV fleets which include Amazon's Kiva robots) to exchange data with production line agents (conveyor belt control units) at smart fulfillment centers run by Amazon and JD.com. The system initiates AGV agent dialogues based on real-time load data which results in a 5-minute pause for material transport to prevent workflow interruptions and decrease AGV system downtime by 10-15%. The system requires agents to respond immediately when environmental conditions change (reactivity) and to start actions which will help the system reach its objectives (proactivity) for PdM to transition from "passive repair" to "active prediction" [5]. The maintenance agents at TSMC's 3nm Fab plants and other semiconductor wafer manufacturing facilities use MAS to combine two different datasets. The system uses real-time sensor data to shut down equipment instantly when etching chamber temperature reaches 5°C above normal to prevent wafer damage (reactivity). The system uses six months of degradation information to schedule maintenance operations in advance which results in a 20% decrease of unexpected equipment shutdowns [27]. *Interoperability*: MAS enable seamless integration across heterogeneous systems, such as legacy machinery and modern IoT devices, by leveraging standardized interfaces and protocols to ensure agents from diverse platforms can communicate effectively [13]. The characteristic serves as a critical element for Industry 4.0 because it enables agents to operate in brownfield environments which contain a combination of legacy equipment with modern sensors that require no special modifications for data exchange [14]. Smart factories use middleware solutions which include OPC UA and DDS to enable agent communication through protocol translation of Modbus to MQTT for real-time coordination between legacy PLCs and cloud-based analytics [14]. The method implementation has decreased system integration expenses by 15–25% in automotive manufacturing facilities [11]. The implementation of systems which use equipment from different manufacturers faces ongoing challenges because of communication protocol differences and meaning discrepancies which result in system performance issues and errors [13] thus requiring adaptive

ontologies or federated standards to boost system reliability [19]. The system achieves fault tolerance through MAS which implement redundant agents and distributed decision-making systems [21]. The system remains vulnerable to security threats because cyberattacks which use adversarial inputs can shut down its predictive agents [20]. Federated learning approaches mitigate privacy risks in shared-data MAS for maintenance [19]. The current research demonstrates that MAS systems execute PdM operations because they have been deployed across different operational environments. Organizations achieve a 15% reduction in maintenance costs through the implementation of MAS agents which schedule wind turbine maintenance operations [10]. The integration of AI technology with MAS systems in petrochemical facilities allows them to perform failure prediction using data from multiple sensors [12]. The system advances in MAS do not address human operator integration because it views operators as outside entities instead of working together as part of the system [3, 23].

3) *Case Studies and Current Industrial Applications*: To illustrate these characteristics, Table 1 presents selected case studies from recent MAS implementations in industrial settings.

These examples demonstrate MAS' potential for efficiency gains but underscore the need for enhanced human collaboration, as explored in subsequent sections.

B. Predictive Maintenance with Machine Learning and Reinforcement Learning

The PdM predictive maintenance system uses data analytics to predict when the equipment will fail, helping manufacturing operations reduce equipment downtime and expenses [5]. The PdM method based on traditional statistics has received enhancement through machine learning (ML) and reinforcement learning (RL) which enable better predictive maintenance strategies [6]. The predictive maintenance system based on ML technology employs supervised learning methods to identify faults and regression models for predicting the remaining operational period of assets [5]. The ML models in [11] achieved 95% accuracy for robotic arm anomaly detection which happens during automotive assembly line operations. Deep learning variants which include convolutional neural networks process sensor information to detect initial signs of wear in semiconductor etching tools [27]. ML models function as reactive systems because they need to learn from past data patterns to make decisions but they cannot optimize scheduling operations in real-time [6]. RL solves this problem through Markov decision process modeling of PdM which enables agents to discover the best maintenance strategies by receiving rewards [16]. Multi-agent RL (MARL) allows the control of big fleets which need to operate under changing environmental conditions like wind turbines [16]. The RL system in manufacturing operations reduces expenses through its ability to determine optimal maintenance schedules which maximize production output [15] while achieving better results than heuristic methods by 20–30% [26]. The RL-based PdM

system lacks ability to handle system uncertainties because it depends on autonomous control for all operations [3]. The implementation of federated PdM systems faces two main obstacles which stem from data privacy concerns and the risk of adversarial attacks [19] [20]. The predictive abilities of PdM become more accurate through ML/RL but the implementation of MAS for distributed execution operations faces limited integration [12].

C. Human-in-the-Loop systems in Predictive Maintenance

HITL systems use operator knowledge to enhance AI decision-making capabilities which overcome PdM system weaknesses when dealing with unclear or unfamiliar situations [7]. The HITL frameworks implement feedback loops which enable model adaptation to achieve a 30% decrease in manufacturing false positive errors [23]. XAI agents in explainable AI (XAI) systems generate predictions which operators can understand to verify automotive robotics maintenance alerts [28]. The HITL system in MAS environments allows human operators to take control of AI decisions when the system faces situations where it cannot make accurate decisions [25]. Digital twins allow human operators to study maintenance simulation results which they use to improve their operational abilities [24, 30]. The real-time dialogue functionality of Large language models (LLMs) allows operators to execute multi-agent PdM tasks because they can access the system right away [31]. The petrochemical facility maintenance operations use federated HITL to protect privacy rights because they need human feedback for their operations [19]. The system of HITL functions without defined communication protocols which restricts human operators to only monitor system activities [3, 4]. The research field of HITL-MAS cyberattack resilience has not received sufficient investigation [21].

D. Identified Gaps and Motivation

The system maintains multiple operational gaps which include (i) MAS systems focus on achieving autonomy instead of working with human operators [3, 23] and (ii) PdM systems that use ML/RL for scheduling only respond to situations while they fail to schedule operations in advance for multiple performance targets [6, 15] and (iii) HITL systems do not fully investigate how systems should work together in unpredictable situations [4, 7]. The combination of different systems in heterogeneous MAS systems creates scalability problems because of interoperability issues which researchers [13, 14] have identified. The current systems for data-driven Predictive Maintenance (PdM) lack proper solutions to protect user privacy and maintain system security according to [18, 20]. The current research fails to establish a single framework which links MAS decentralization to PdM forecasting and HITL judgment for developing human-oriented maintenance systems which defend system stability [25, 26]. The IAPM framework exists to solve these problems by uniting hybrid ABMS-RL with human collaboration which starts when uncertainty occurs to create Industry 4.0 PdM systems that adapt and maintain trustworthiness.

This table underscores the fragmented nature of prior work, with few integrating MAS, RL, and HITL for proactive, human-centric PdM.

III. PROPOSED IAPM FRAMEWORK

The IAPM framework presents an innovative hybrid system which unites Multi-Agent Systems (MAS) with Reinforcement Learning (RL) and Human-in-the-Loop (HITL) mechanisms to achieve proactive predictive maintenance (PdM) with Industry 4.0 resilience and human-centered operations. The system allows human operators to actively participate in the MAS which converts autonomous decision systems into adaptive collaborative systems that respond to manufacturing environment changes. The framework consists of six stages which function as a unified system to merge artificial intelligence processing with human expert knowledge. The following overview explains how these phases organize their structure and their connections but Section 5 contains the full details for their execution.

A. Modeling and Agent Definition

The IAPM system operates through a decentralized system which uses Agent-Based Modeling and Simulation (ABMS) principles to create its foundation. The main entities and their communication methods become defined during this stage to create distributed intelligence which includes human participation. The framework consists of four main agents which operate through an ABMS system that serves as their coordination mechanism. • The Machine/IoT Agent functions as the physical asset (e.g., resistance spot welding machine) and handles real-time sensor data streaming. • The system depends on AI Predictive Agent to analyze data for failure prediction and severity score calculation through machine learning models. • The Human Operator Agent operates as a system which provides users with interactive dashboard access to contextual information for performing overrides and situation refinement when needed. • The RL Scheduling Agent determines maintenance schedules by analyzing multiple performance indicators which it can process. The system operates through bidirectional communication protocols which use MQTT-inspired messaging to enable agents to exchange data while they obtain feedback from each other. Table 1 presents agent functions together with their corresponding system-enabling components. Figure 2 illustrates the detailed multi-agent architecture. Figure 2: Detailed Multi-Agent Architecture of the IAPM Framework. (Caption: The architecture shows interactions among agents, with bidirectional arrows for communication and the ABMS layer for coordination.)

B. Real-Time Data Integration

The system develops an efficient data processing system which streams operational data into the MAS at high speed for immediate PdM activation. The spot welding machine generates sensor data which gets transformed into health indicators through electrode force and current measurements and ultrasonic score and tip temperature and robot error and vibration

RMS data processing. The pipeline uses MQTT or Kafka technologies to achieve high-throughput streaming data from PLCs/sensors which runs through fault-tolerant systems to edge/cloud platforms. The process of data validation requires three steps which include missing value handling through median imputation and outlier removal and timestamp synchronization for all data points. The 100,000-sample dataset maintained its original time sequence because researchers conducted data analysis by splitting the data through chronological methods. The system allows for immediate feature processing which feeds into the AI Predictive Agent.

C. Fault Prediction and Human-AI Collaboration

Organizations at this stage need to create exact failure prediction systems which must operate with team-based improvement approaches to manage unexpected events. The AI Predictive Agent uses combined data to create maintenance urgency ratings and severity levels which deliver statistical failure risk information. The Human Operator Agent reviews all predictions which need evaluation when the system identifies situations that remain unclear through its confidence threshold analysis. Operators can modify system outputs through their understanding of environmental conditions which sensors fail to detect which results in a collaborative system that improves both precision and system reliability. The joint effort between these organizations helps to decrease false alarm occurrences while making AI system results match the actual operational environment. Figure 4: Human Dashboard Mockup. (Caption: The interface displays AI predictions, severity, suggested actions, uncertainty indicators, and override options for collaborative refinement.)

4.4 Proactive Maintenance Scheduling The system phase establishes the time for intervention implementation which defends systems from breakdown while maintaining operational continuity through human participation and various performance metrics. The RL Scheduling Agent performs evaluation of predicted outcomes together with severity levels and available resources and production capacity and operational restrictions to choose between scheduling immediately and postponing the task. The system achieves risk management and downtime reduction and cost optimization through its adaptive policies which human operators verify for feasibility during critical or unpredictable situations. The method allows PdM operations to transition from their current reactive emergency response model into strategic maintenance which supports operational goals. Figure 3: Workflow Sequence Diagram of the IAPM Framework. (Caption: Sequential flow from data integration through prediction, human collaboration, RL scheduling, and feedback. Dashed lines show optional overrides.)

D. Continuous Monitoring and Feedback Loops

The system creates active loops which enable continuous adaptation and performance enhancement during this stage. The system tracks all post-maintenance results together with system performance data and human interaction records. The

collected data enables the system to perform sequential improvements of its predictive models together with its scheduling mechanisms and ABMS rules which results in a system that adapts by itself. The system improves its rules through pattern recognition of override and failure data and feedback aggregation which enables the AI system to achieve better accuracy. The system maintains long-term stability through its closed-loop operation which will achieve complete real-time functionality during upcoming deployments to address changing operational requirements.

E. Validation and Analysis

The framework evaluation at this stage uses simulation-based metrics to check its performance while keeping its goals in line with Industry 4.0 objectives. The main performance indicators include prediction accuracy and failure prevention rates and downtime/cost trade-offs and collaboration impacts. The sensitivity analysis evaluates how different parameter values affect the system because it checks how changes in uncertainty thresholds and data noise levels impact the results. The framework shows promise because Section 6 of the results presents no system failures during the initial simulation tests and the system maintains its operational performance at the same level. The research phase produced particular findings which helped researchers enhance their work between theoretical design and experimental testing.

F. Key Contributions and Innovations

The IAPM framework brings two innovative elements which allow for dual communication methods and human participation through uncertainty detection to establish a human-AI partnership that extends past conventional AI-based PdM systems. The system operates differently from conventional MAS because it includes operators who function as active participants who can perform immediate system intervention when the system produces uncertain results at medium severity levels. The uncertainty mechanism which uses score thresholds to measure uncertainty enables decisions to combine computational accuracy with human judgment while solving the problems identified in [7, 23]. The system depends on ABMS to achieve stable coordination between system elements while RL performs dynamic optimization to achieve multi-objective equilibrium between risk and cost and downtime which previous studies [15, 16] were unable to achieve.

IV. METHODOLOGY